

Combining Quantum with Physics-Inspired AI for Accelerating CFD Sim- ulations

From PINNs and neural operators to hybrid quantum–classical
models

A Seshaditya

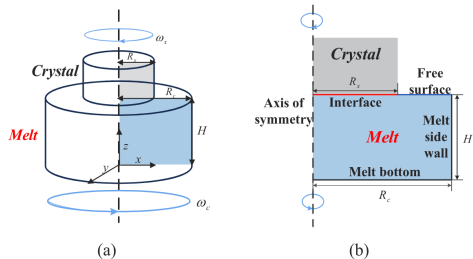
QuaSi

The problem: CFD is accurate, but expensive

Why accelerate?

- ▶ High computational cost of high-fidelity CFD.
- ▶ Expensive parameter sweeps and repeated design iterations.
- ▶ Multi-physics coupling increases model and training complexity.

Goal: learn useful solution structure while retaining the governing physics.

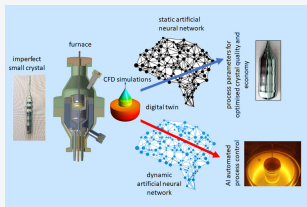


Source: Silicon crystal-growth process schematic; source material in supplied deck.

Application focus: two demanding CFD settings

Silicon crystal growth

- ▶ Czochralski (CZ) process.
- ▶ Coupled Navier–Stokes, particle transport and heat transfer.
- ▶ Strong thermal–fluid interaction.

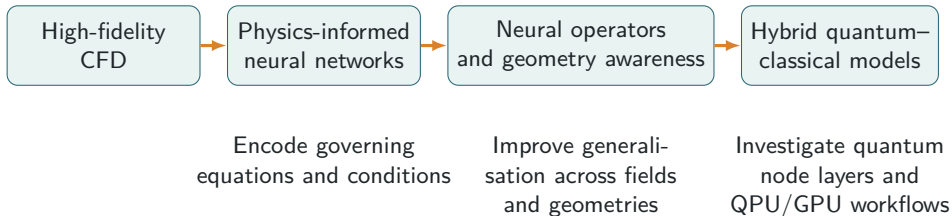


Cryogenic / hydrogen systems

- ▶ Design-oriented CFD for complex flow and thermal conditions.
- ▶ Faster surrogate models can support iterative design.
- ▶ Relevant to the broader space and energy use cases in the source deck.



The technical story



Research question: where can hybrid quantum-classical learning add value without discarding the physics structure?

Physics-Informed Neural Networks (PINNs)

Core idea

A neural network approximates the solution field while the loss function penalises violations of the governing PDEs and boundary/initial conditions.

- ▶ Less dependence on large labelled datasets.
- ▶ Natural fit for multi-physics equations.
- ▶ Automatic differentiation provides PDE residuals.

Source: <https://doi.org/10.1063/5.0226232>

PINNs

- Solve PDE of the form
 $\mathcal{L}(\mathbf{u}) = \mathbf{0}$ on domain Ω with BCs and ICs.
- Define trial solution $u_\theta(x)$.
- Minimize loss:

$$\mathcal{L} = \lambda_f \|\mathcal{L}(u_\theta)\|_{\Omega_f}^2 + \lambda_b \|u_\theta - g\|_{\partial\Omega}^2 + \lambda_i \|u_\theta - u_0\|_{t=0}^2$$

Essentially: They optimize a differentiable surrogate that satisfies the PDE.

PINN methodology: physics becomes part of the objective

$$L_{\text{total}} = L_{\text{PDE}} + L_{\text{BC}} + L_{\text{IC}} + \lambda L_{\text{SAL}}$$

L_{PDE} physics-informed residual.

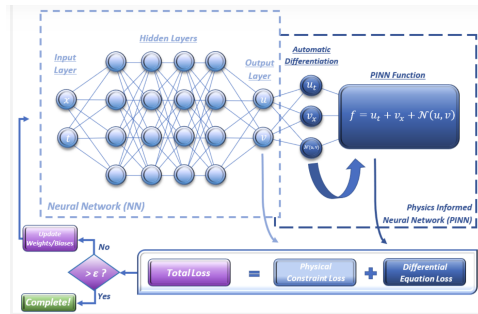
L_{BC} boundary-condition loss.

L_{IC} initial-condition loss.

L_{SAL} sensitivity-analysis term.

λ weighting factor.

Computation: automated differentiation.

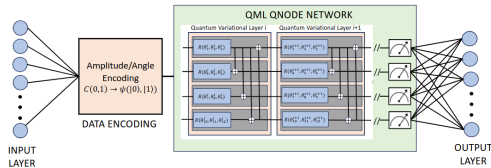


From PINNs to neural operators

Why move beyond a single solution?

Neural operators learn mappings between functions rather than only one fixed field. This can improve reuse across parameter settings and related problems.

- ▶ PINO: physics-inspired neural operators.
- ▶ Operator learning targets families of PDE solutions.
- ▶ The source deck emphasises higher accuracy, multi-physics capability and data efficiency.



Source: PINO/physics-informed operator illustration from supplied deck.

Geometry-aware operators: the next generalisation step

PI-GANO

Geometry-aware neural operators aim to generalise across PDE parameter representations and geometries.

Source: arXiv:2408.01600v1

GAOT / MAGNO

The deck also introduces Geometry Aware Operator Transformers and multiscale attentional graph neural operators.

Source: arXiv:2505.18781v2

★ Why Use PINO Instead of PINNs?

PINO combines the advantages of both:

Feature	PINO	PINN
Learns mappings between <i>functions</i>	✓	✗
Good for fast prediction on new inputs	✓	✗
Uses physics constraints	✓	✓
Data-efficiency	High	Medium
Computational cost during inference	Very low	High

A compact comparison of the classical approaches

Approach	Primary idea	Strength in the source deck	Open challenge
PINN	NN + PDE residuals	Physics constraints, data efficiency	Training stability and scaling
PINO	Operator learning + physics	Families of PDE solutions	Generalisation across settings
PI-GANO	Geometry-aware operator	Geometry generalisation	Complex geometries / representations
GAOT	Geometry-aware transformer	Attention-based operator learning	Computational cost and validation

Aspect	PI-GANO	GAOT (e.g., GINO)
Physics-Informed	Yes (can train without labeled data)	No (typically requires labeled simulation data)
Geometry Handling	Explicit geometry encoder, geometry-aware	SDF and point-cloud encoding, geometry-aware
Data Efficiency	High (physics-informed, less data needed)	Lower (relies on simulation data for training)
Generalization	Across geometry and PDE parameters	Across geometry (parameters less emphasized)
Computational Speed	Fast (neural operator-based)	Extremely fast (demonstrated 26,000x speed-up)

Case study: silicon crystal growth

Czochralski (CZ) process

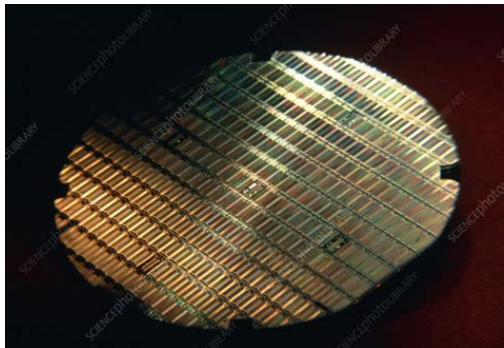
The supplied material frames the problem as a coupled thermal–fluid system involving:

- ▶ Navier–Stokes equations,
- ▶ particle transport,
- ▶ heat transfer.

Source: <https://doi.org/10.1063/5.0271778>

PINN role

Parameter sensitivity is highlighted for temperature gradients, precursor pressure/concentrations, flow rates and substrate properties.



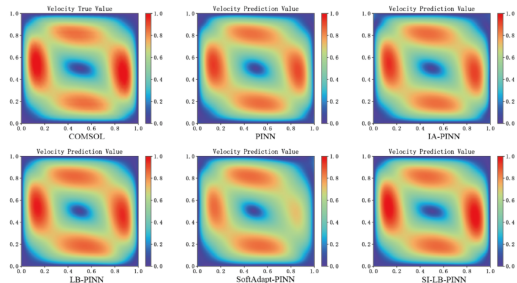
Improving the PINN for thermal–fluid coupling

Spatial information + adaptive loss balance

The source deck describes an approach intended to:

- ▶ optimise thermal–fluid coupling,
- ▶ improve training stability,
- ▶ balance contributions from different physics terms.

Source: <https://doi.org/10.1063/5.0271778>



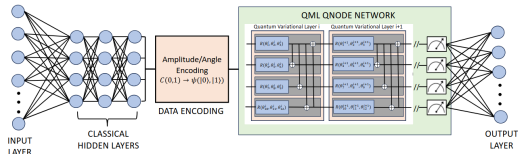
Representative simulation/result material from the supplied presentation.

Why bring quantum computing into the workflow?

Hybrid rather than replacement

The proposed direction is to combine classical physics-informed models with quantum components, using QPU/GPU hardware where appropriate.

- ▶ Quantum node layers inside otherwise classical architectures.
- ▶ Variational quantum circuits (VQCs).
- ▶ Exploration of quantum neural operators and quantum-enhanced PINNs.



Source: Hybrid quantum–classical PINN architecture; source:

arXiv:2304.11247v3.

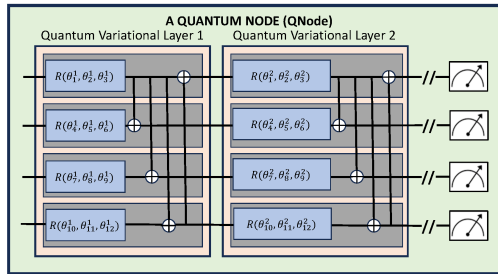
Quantum PINNs: candidate architectures

Quantum node layer

A quantum layer can be embedded into a classical neural architecture, creating a hybrid model rather than a fully quantum solver.

- ▶ Multi-variational quantum layers.
- ▶ Classical layers surrounding quantum components.
- ▶ Geometry-focused VQCs.

Source: <https://www.mdpi.com/1099-4300/26/8/649>



Source: Quantum-node / variational-circuit architecture from supplied deck.

Quantum operators and attention-enhanced models

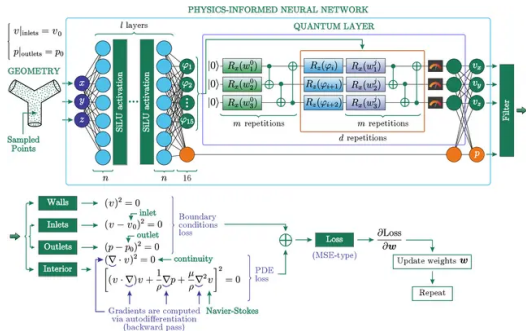
Quantum DeepONets

The deck points to quantum-enhanced operator learning as a route toward neural operators accelerated by quantum computing.

Source: <https://arxiv.org/pdf/2409.15683>

Attention-enhanced QPINNs

Attention mechanisms and tensor-network ideas are also presented as candidate routes for improving expressive capacity.



Where quantum advantage is—and is not—established

Important distinction

The supplied presentation treats quantum advantage as a **research objective**, not as an established result for general CFD.

What is demonstrated or proposed

- ▶ Hybrid quantum/classical architectures.
- ▶ Quantum node layers and VQCs.
- ▶ Candidate PDE and CFD applications.
- ▶ Reduced training-data requirements are discussed for specific case studies.

What still needs benchmarking

- ▶ Wall-clock time versus strong classical baselines.
- ▶ End-to-end cost including QPU access and data movement.
- ▶ Accuracy across geometry and parameter changes.
- ▶ Reproducibility on realistic CFD workloads.

Source: QCPINN case study: [arXiv:2503.16678](https://arxiv.org/abs/2503.16678); supplied deck notes a potential 80–90% reduction in training volume for that study

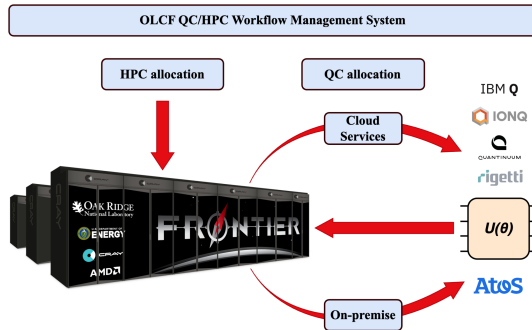
Implementation: connect QPU and HPC/GPU resources

Software / hardware ecosystem shown in the deck

- ▶ PennyLane
- ▶ NVIDIA Modulus
- ▶ DGX Quantum
- ▶ IBM Quantum

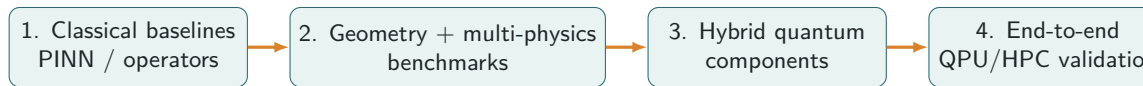
Workflow principle

Keep the expensive, highly parallel CFD and neural-network operations on classical HPC/GPU infrastructure, while testing quantum components where they can be isolated and benchmarked.



Source: QC-HPC integration architecture; source:
arXiv:2408.16159v1.

A practical research and validation roadmap



Benchmark dimensions

- ▶ Accuracy and conservation properties.
- ▶ Training/inference time.
- ▶ Data and memory requirements.

Validation settings

- ▶ Silicon crystal growth.
- ▶ Turbulent-flow examples.
- ▶ Cryogenic / CFRP vessel examples referenced in the source deck.

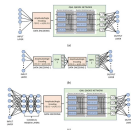
Outlook: from promising components to validated CFD workflows

Near-term

- ▶ Benchmark physics- and geometry-aware AI models.
- ▶ Compare hybrid quantum models against strong classical baselines.
- ▶ Quantify where quantum components change the accuracy/cost trade-off.

Longer-term

- ▶ QPU/GPU integrated simulation pipelines.
- ▶ Academic and industrial collaborations.
- ▶ Pilot projects for aerospace and related engineering applications.



Core message: physics-informed learning provides the foundation; geometry-aware operators improve reuse; hybrid quantum models are the next hypothesis to test.

Key takeaways

1. **CFD acceleration is the driver:** high-fidelity multi-physics simulations are computationally expensive.
2. **PINNs add physics constraints:** PDEs, boundary conditions and initial conditions become part of learning.
3. **Operators add generalisation:** PINO, PI-GANO and GAOT target mappings across solution families and geometries.
4. **Quantum is a hybrid research direction:** QPINNs, quantum node layers and quantum operators can be tested within classical workflows.
5. **Validation is essential:** claims of quantum advantage need end-to-end, problem-specific benchmarking.

Physics + geometry + hybrid quantum resources
→ a coherent path toward accelerated CFD

Selected references from the supplied deck

- ▶ Afrah Farea, Saiful Khan, Mustafa Serdar Celebi (2025), *QCPINN: Quantum-Classical Physics-Informed Neural Networks for Solving PDEs*, arXiv:2503.16678.
- ▶ *Hybrid Quantum PINNs for Computational Fluid Dynamics* (2025), IOP Science, doi:10.1088/2632-2153/ad43b2.
- ▶ *Quantum Physics-Informed Machine Learning for Multiscale Simulations* (2024), arXiv:2403.08954.
- ▶ *PINNSFORMER: A Transformer-Based Framework for Physics-Informed Neural Networks* (2023), arXiv:2307.11833v3.
- ▶ Pengpeng Xiao et al. (2024), *Quantum DeepONet: Neural Operators Accelerated by Quantum Computing*, manuscript.
- ▶ Siddhant Dutta et al. (2024), *AQ-PINNs: Attention-Enhanced Quantum Physics-Informed Neural Networks for Carbon-Efficient Climate Modeling*, arXiv:2409.01626.
- ▶ D. Kolberg, C. Anders, S. Reke (2023), silicon VMCz crystal-growth PINNs, doi:10.1080/00219592.2023.2236656.
- ▶ J. Ling, Y. Zhang, D. Chen (2023), thermal-fluid coupling in silicon crystal growth, doi:10.1063/5.0123811.
- ▶ Y. Zhao, H. Liu, J. Xu (2025), thermal-fluid coupling in Czochralski silicon growth, doi:10.1063/5.0271778.

Thank you

A Seshaditya

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github.com/adytiaa/quasi.ai

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